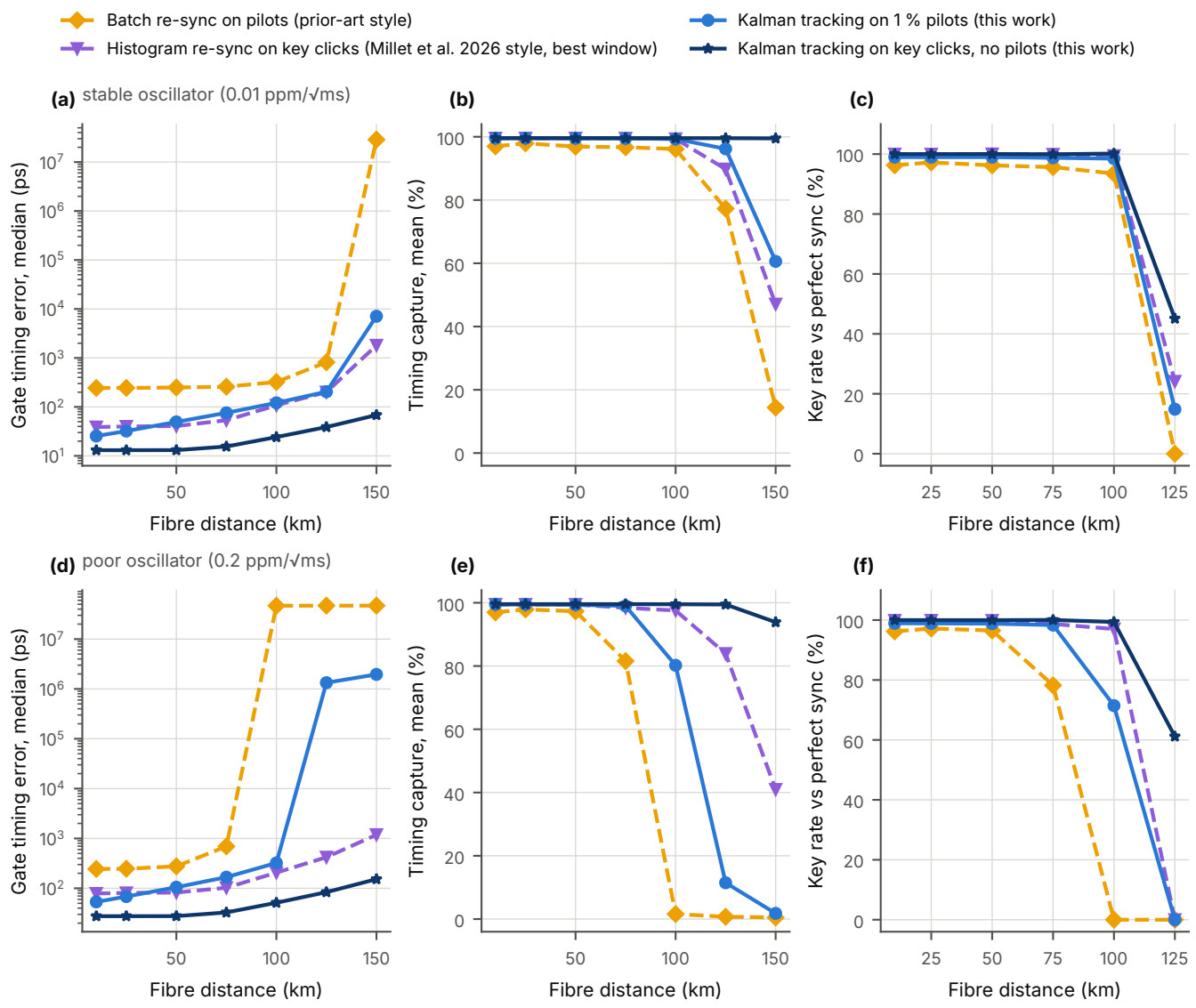
Batch re-sync on pilots (prior-art style)
Histogram re-sync on key clicks (Millet et al. 2026 style, best window)
Kalman tracking on 1 % pilots (this work)
Kalman tracking on key clicks, no pilots (this work)
(a) stable oscillator (0.01 ppm/√ms)
(b)
(c)
(d) poor oscillator (0.2 ppm/√ms)
(e)
(f)
Gate timing error, median (ps)
Timing capture, mean (%)
Key rate vs perfect sync (%)
Fibre distance (km)